Towards Audio-Based Music Information Retrieval in the RMIT MIRT Project

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Abstract

Musical Information Retrieval is the field of research that explores ways of retrieving music, mainly using non-text techniques, such as through melody queries or rhythm. At RMIT Computer Science we have worked on melody-based information retrieval of MIDI files and are now exploring the use of audio features, such as rhythm, tempo, tonality, noisiness, and timbre to answer a range of music queries. In this paper we summarise the achievements of the MIRT project, and describe our current projects. In addition we show some preliminary work that explores the application of thresholding on standard deviation of frequency spectrum values to the analysis of cymbal hits.

1 Introduction

Traditional textual databases of musical information don't allow certain types of music query that are commonly asked, such as, "where does this fragment of melody come from?", "who wrote the Alleluia that goes like this?", "how similar is my composition to existing works?", "I like the song 'I've seen that face before'; what else is there that is similar?", "I want a piece of music for my film, that expresses defiance". The field of Music Information Retrieval aims to address these types of content-based queries.

RMIT's MIRT project commenced in January 1997 as work for a PhD thesis (Uitdenbogerd 2002). The initial aim was to develop techniques that allow a melody query to be presented to a system, which would then provide a ranked list of matches to the query based on melodic similarity. This work led to the development of a three stage approach to music matching, consisting of melody extraction (TRIMming), melody standardisation (TIRMing), and similarity measurement (MITRing). These three stages work together to ensure that meaningful matches can be found to melody queries, given the processing constraints and varied musical content associated with matching against a large collection. The work focused on musical data in the MIDI file format and searching polyphonic music for matches to melody queries. Techniques that worked well for this problem were the extraction of melodies based on the highest pitch note commencing at each instant in each MIDI track, representing these melodies as strings of pitch differences, and using a matching method that counts all common substrings of length 5 (n-grams), ignoring duplicates (For more details see Uitdenbogerd and Zobel (2002)).

While we are continuing to answer the research questions associated with the melody matching problem, we have expanded our project to include other music retrieval applications. Methods of retrieving music that are likely to be useful include those based on style, mood, and musical taste. In order to address these types of queries, we are exploring methods of extracting the features from music in audio format that may indicate style or mood. We will be combining features that are already published, such as those used by (Welsh et al. 1999), and Tzanetakis, Essl, and Cook (2001), with others that try to capture aspects that have been shown to be important through studies of human perception of music. Early work in music psychology suggests that for determining the mood of a piece, tempo is the most important, followed by tonality and then whether the pitch is high or low (discussed in Farnsworth (1958) Chapter 5). For musical style, human-classified genres have been based on genealogical, historical, geographical, functional, instrumental, as well as other enumerated types (Pachet and Cazaly 2000).

For musical taste, we expect to answer queries based on a user's existing music preferences, that is, we are developing techniques for music recommender systems. An approach that has been successfully applied to this type of problem is known as collaborative filtering. We are building a test platform for evaluating different methods of recommending music. These will include user ratings and behaviour in addition to other forms of evidence, such as the answers provided to questionnaires presented to the user, and user-provided links between pieces.

Another problem that has yet to be solved is how the lay person can reliably find music for which they know a part of the tune. Parsons (1975) used a melody "contour" index of themes, that is, an alphabetically sorted list of themes represented by a sequence of symbols representing *up*, *down*, and *same* note transitions.

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This made it possible to look up a theme in the index without being able to read music. Some prototypes have been developed that allow users to sing a melody query, a melody contour is then extracted and used to look up answers in a database (Blackburn and Roure 1998) (Ghias et al. 1995; McNab et al. 1996). It has been found, however, that the use of contour information is insufficient for larger collections of musical data (Uitdenbogerd and Zobel 1999).

Beyond the question of music representation, it is still unclear how to best present the answers to a music query so that the user can determine whether they are indeed relevant. We concluded from an experiment in collecting music relevance judgements that it is important to provide match location information when the user wants to listen to the potential matches. There are further difficulties for users when listening to unfamiliar music (Uitdenbogerd, Chattaraj, and Zobel). In work that extends our earlier melody-based retrieval research, we are exploring different types of music retrieval user interface, and evaluating them from a usability perspective. Our initial experiment will compare the usability of different textual representations of melody for novice users.

In other work, we hope to examine the playing style of musicians using various audio analysis techniques. Initially we are focusing on percussion style, which should also be a useful indicator of genre. Percussion playing style can be characterised by timing, loudness, and timbral variation of hits. Using existing techniques for locating beats, we can explore timing variations in pieces of music. However, these techniques do not isolate percussion from other strong beats occurring in the music. As an indicator of percussion rhythm it may be interesting to isolate the high-hat and other cymbals that are used for fast beats. As a preliminary step we are analysing audio samples of high-hat and cymbal hits using Fourier Transforms and statistical approaches, to determine the nature of a cymbal attack. This preliminary work is reported in this paper. Other researchers have published on related topics, such as instrument detection (Fujinaga and MacMillan 2000; Martin and Kim 1998), beat detection (Cemgil et al. 2000) (Dixon 2001), transcription of music (Garcia 2001; Hainsworth and Macleod 2001; Kashino and Tanaka 1993; Martin 1996a; Martin 1996b), and the analysis of playing styles (Cambouropoulos 2001; Trilsbeek, Desain, and Honing 2001). To our knowledge, however there hasn't been much work specifically on percussion analysis.

2 Background

In the preliminary work we report here we used Fourier transforms of a stored wave-form. In this section we provide a brief review of the relevant concepts.

When audio is stored on a computer, the sound signal is converted from a continuous one to a discrete one via analogue to digital conversion. Each sample represents an amplitude at a specific time unit. While it is possible to glean some useful information from audio stored in this manner, more meaningful details are able to be extracted if the data is transformed into another representation. The Fourier transform is one such transformation.

The Fourier transform takes a signal such as one from an audio source, and converts it from a function of time to one on frequency. It is based on the Fourier series, which can be used to represent a periodic signal in terms of an infinite weighted sum of sines and cosines. A finite signal that is not periodic can be treated as periodic by assuming that the period of the signal is equal to the duration of the entire signal. In practice, when applying these techniques to discrete audio signals, the transform usually provides frequency bins of a particular range and the magnitudes associated with them. For detailed analysis of changes in the frequency spectrum over time, short sequences of audio are transformed, allowing a sequence of frequency snapshots to be compared. These can be analysed statistically.

When examining the timbre of an instrument, the frequency range, or *bandwidth*, is an important indication. Much of the frequency spectrum for a particular timbre remains the same regardless of the pitch of the base note (Dodge and Jerse 1997). Conversely, if the frequencies shift with the base note, the timbre appears to change. For example, consider the effect of changing the pitch and speed of a voice sample.

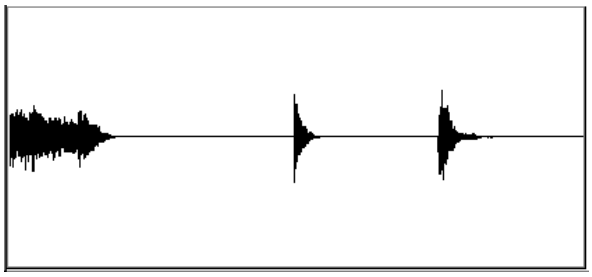


Figure 1: The wave-form used for our preliminary work. It consists of a high-hat being closed, and two crash cymbal hits.

3 Preliminary Work

The goal of this initial project was to establish whether a line of investigation into the possibilities of profiling timbres according to levels of volatility or variation was promising enough to pursue. The idea came about while viewing the three-dimensional spectrograms of many instruments and their unique timbres. It was a view of the profile of percussion instruments, especially various crash cymbals, that highlighted the magnitude of noise propagated across the frequency spectrum for the duration of the cymbal's decay. The

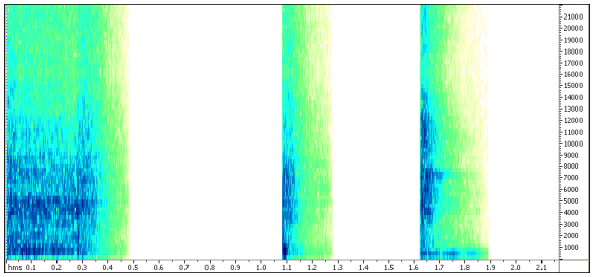


Figure 2: The power spectrum of the wave-form shown in Figure 1. Observe the spread of frequencies (bandwidth).

term noise is being used in the sense of not having the clear fundamental and regular harmonic properties produced when playing most pitched instruments. A clarinet, for example, will have a strong fundamental frequency and some overtones (harmonics) which characterise the timbre as being that of the clarinet. The degree of noise produced when sounding particular instruments could well disrupt an audio analysis model. If it were possible to do a probabilistic model, assigning a particular noise interference profile to a particular event within the sample, it may be possible to compensate for this in an analysis, such as focusing on the area in question and using statistical techniques to remove or reduce the effect of the increased variation, thereby enabling a finer analysis of what may be masked by such noise. The scope of such techniques was well beyond the limits of this initial investigation, which was limited to determining the effectiveness of such an approach to timbres in a non-polyphonic setting.

3.1 Method

The aim was to determine what kind of timbral profiles would be returned when assessing a sound in relation to an average measure of amplitude variation, in a particular frequency spectrum. The two main variables were the length of the time average and the level of amplitude extremity (or distributions falling within certain deviations away from the average variation). Variation in the time domain was taken to mean "Is the current amplitude of this frequency bin, greater or lesser than the average of an arbitrary amount of consecutive previous frequency bins in the same spectrum?" The level of extremity variable means "Is the current amplitude of the frequency bin greater than N standard deviations away from the average of the amplitudes of the same frequency over an arbitrary distance?" Combinations of these results had some interesting effects.

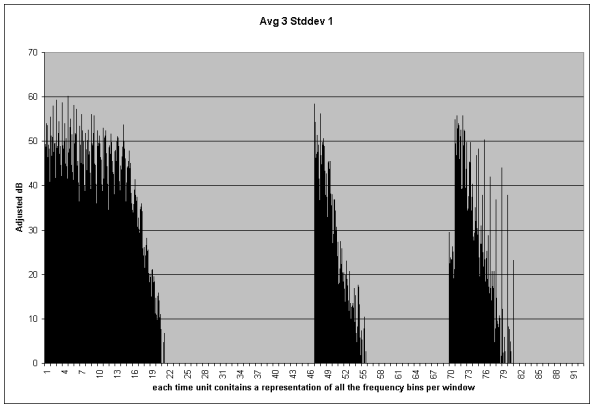
Since the idea came from viewing crash cymbals, we commenced the investigation by creating a small library of different types of cymbals with differing timbral profiles and sequencing them into a wave file (see Figures 1 and 2). This enabled the extraction of the frequency profiles by using the Fast Fourier Transform (FFT). After some experimenting with various lengths, the use of a window length of 1024 and FFT length of 1024 was selected, as it seemed the most appropriate for this current analysis, in that percussion was the focus and therefore accuracy in the time domain was more important than accuracy in the pitch domain. Higher window values were tested but the resolution in the time domain was reduced such that the data returned was not accurate enough to provide the detail required for this style of analysis.

3.2 Observations

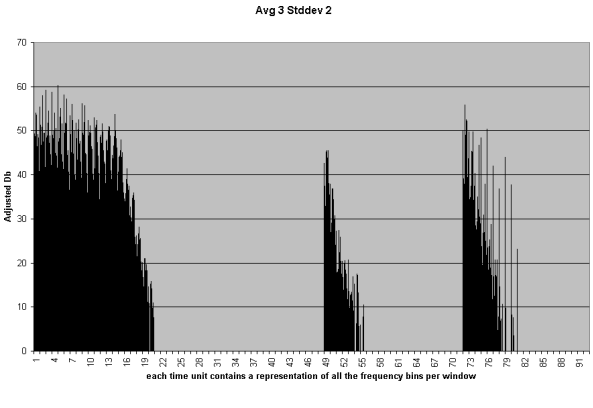
Figures 3, 4, and 5 show the values for different averaging windows and thresholds. The horizontal axis shows two dimensions: each unit represents a time unit, and within these are vertical lines representing the amplitude for each specific frequency band.

In the cases of the average of 3 time units, there is not a great deal of change between the different deviations (See Figure 3). When the average is increased to 6, the change is more immediate. The first plot (6,1) (See Figure 4) is not much different from all the previous ones but the next, (6,2) displays a severe curtailing of the decay. This is to be expected as the effects of the initial amplitude spike has "washed" through the calculations. The final plot of the average of 6 (6,3) shows the other effect of having such an amplitude spike; the deviation from the mean is so great, that when using a factor of 3 in the calculations, very little else exceeds it until there is great calm in the music, a situation not obtained in this short sample. The second and third cymbal hits do not even register and are absent from the plot entirely. As the average is increased to 9 units (not shown), the effects of altering the sound profiles are more predictable. With a steady increase of the deviation factor, the decays reduce proportionately. As the time factor here is three times greater than in the first instance, the averages will be less reactive to sudden spikes but the effects will remain in the calculations for a longer period. The final set, based around an average of 12 (For example, see Figure 5), display similar properties to 9 but have slight differences in which frequencies are more persistent. It is interesting to note that these differences are only those that could be picked up by the naked eye and it is possible that with more advanced comparison functions, the program might quantify other significant differences, especially within each window, which has been compressed by the nature of the graphical pot. Further development is needed to extract the differences between the models and distil them into meaningful patterns. It is obvious that a profile as generated by such a routine is a degree more descriptive than one that is just looking for a spike above a given level. This also provides data relating key features of the timbre of the sound. The effect is to attempt to create a data set that could be stored for correlation or comparison routines at a

a)



b)



c)

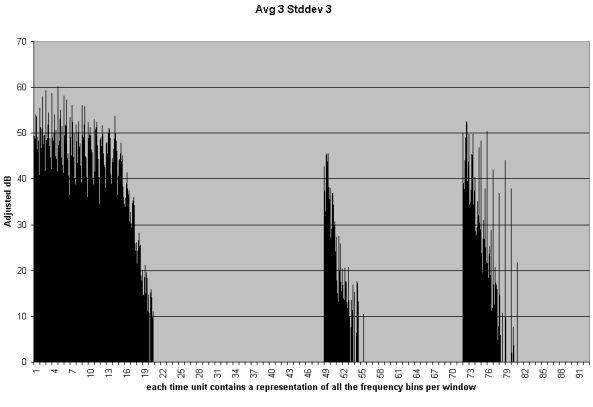
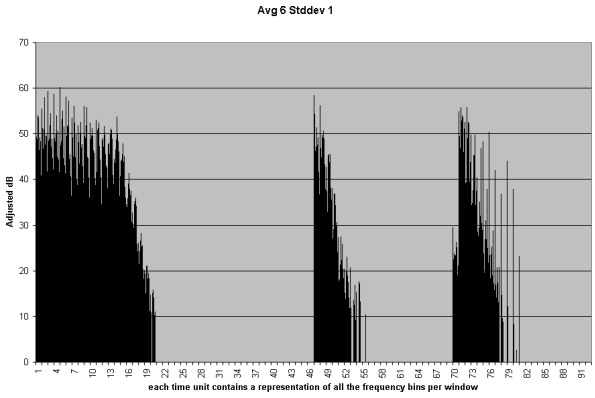
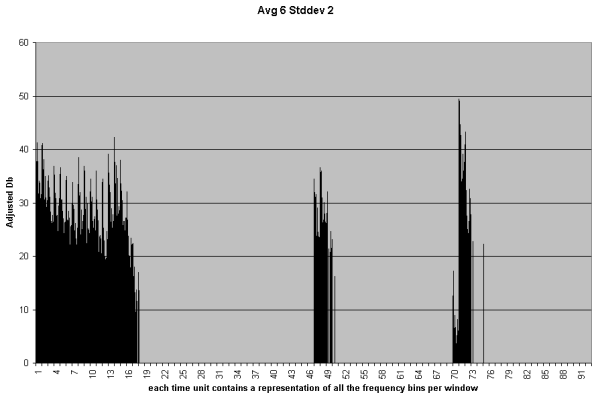


Figure 3: Graphs of the change in frequencies over time. The X axis is divided into bins containing the full frequency domain for a particular FFT window. For these graphs, the current frequency amplitude was compared to the mean of the previous 3 amplitudes. At a), amplitudes that were within one standard deviation were mapped to zero, at b) those within two standard deviations, and at c), within three standard deviations. Values that were less than an arbitrarily selected number, representing low level noise, were also removed.

a)



b)



c)

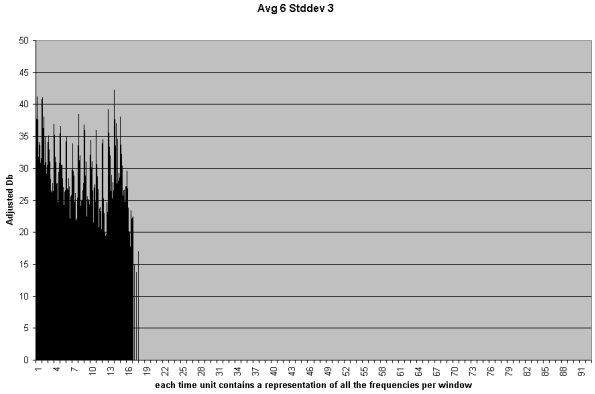


Figure 4: Graphs of the change in frequencies over time. The X axis is divided into bins containing the full frequency domain for a particular FFT window. For these graphs the current frequency amplitude was compared to the mean of the previous 6 amplitudes. At a), amplitudes that were within one standard deviation were mapped to zero, at b) those within two standard deviations, and at c), within three standard deviations.

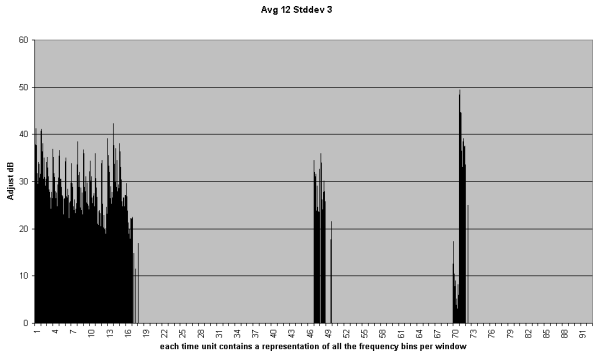


Figure 5: A graph of the change in frequencies over time. The X axis is divided into bins containing the full frequency domain for a particular FFT window. For this graph the current frequency amplitude was compared to the mean of the previous 12 amplitudes. Those that were within three standard deviations were mapped to zero.

later stage.

3.3 Discussion

There are many potential issues surrounding the use of such a technique: how is an optimum average and variation to be arrived at? What relation does this have to the tempo of the music? When would accuracy of frequency outweigh the need for accuracy of pitch within the FFT? It is possible that a number of test runs might be necessary to determine the probability of success of the employment of such an algorithm. It also may have questionable effects when the music is played by pitched instruments which display less variation within a pitch and move across many pitches within a phrase. More testing would be needed to determine a strategy to deal with this.

To conclude, we think that this investigation is worth pursuing. We suspect that any comparison routine based on variation could become quite complex due to the amount of data that would be collected and the number of comparisons to be made. There might exist a trade-off between using a less descriptive average to accommodate more comparisons, and only employing a search using this method of comparison if all others had failed.

4 Summary and Conclusion

There are many interesting research questions related to music information retrieval. Practical systems for melody-based music retrieval from MIDI or note-based data can now be built, but are limited by the data that is available. Most music is available as audio recordings, but extracting meaningful data from audio data is quite difficult. Our MIRT project hopes to address some of the problems that have not yet been solved for audio-based music retrieval. Our focus is on feature extraction of indicators of style or mood, recommendations based on a combination of evidence, and detecting instruments. We are interested in collaborating with other researchers on these ideas.

The preliminary work reported here revealed that a combination of Fourier Transform and the detection of changes in deviation from the mean are reasonably effective in locating the onset, and detecting certain characteristics of a percussion instrument, however, there are many questions that remain unanswered, such as whether the approach will work for more complex audio samples, and whether other types of transform, such as wavelets, will provide this information more readily. We intend to develop these ideas further in the near future.

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